DES646 – AI/ML for Designers

Mid-Term Project Proposal

Project Title: Emotion-Aware Conversational 3D Assistant for Education

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- **Empowering emotionally intelligent virtual learning through AI-driven 3D interaction.**

Project Aim

To design and prototype an AI-powered 3D conversational assistant that can detect, interpret, and respond to students' emotions to enhance engagement and empathy in virtual learning environments.

Objectives

• Develop an emotion recognition model that uses text tones and text cues.
• Integrate the emotion feedback into a conversational AI framework (e.g., LLM + sentiment layer).
• Build an interactive 3D avatar capable of expressive reactions and adaptive dialogue.
• Evaluate its impact on student engagement and emotional comfort during learning.
• Explore ethical and design implications of emotionally aware AI in education.

Proposed Methodology / Technical Approach

1. **Data Collection:** Usage of open datasets such as FER-2013, RAVDESS, and emotion-labelled conversation corpora.
2. **Emotion Recognition Module:** Implementing multimodal classifier using CNNs, speech embeddings, and transformers.
3. **Conversational Layer:** Employ an LLM-based dialogue engine fine-tuned with educational content or using an api key for the base text.
4. **3D Avatar System:** Create a real-time 3D character with emotion-linked animations and TTS.
5. **Evaluation & Testing:** Conduct usability tests measuring engagement and perceived empathy.

Expected Deliverables

Deliverable	Description
Prototype	Working 3D assistant demonstrating emotion-adaptive conversation
Dataset	Curated multimodal emotion dataset (text, speech, face)
Design Documentation	User flow, emotion mapping, and interface design rationale
Presentation	3-min video demo showing emotion-based dialogue adaptation
Final Report	Full documentation with reflections on ethical AI in education

Intellectual Property / Research Dissemination Plan

The team aims to pursue a research paper titled “Emotion-Aware Conversational Agents for Empathetic Learning Environments” for submission to India HCI 2026 or Design Research Society (DRS 2026). Patent potential will also be explored for the multimodal emotion-adaptive interaction framework.

Preliminary Timeline

Week	Activities
1	Literature review + dataset exploration
2	Build baseline emotion recognition models
3	Develop dialogue layer and emotion fusion logic
4	Design 3D avatar and integrate basic emotion visuals
5	Implement real-time emotion-driven responses
6	User testing + design refinement
7	Final demo, documentation, and report submission

Resources Required

- TensorFlow / PyTorch
- OpenCV, DeepFace, librosa
- React + Three.js / Unity
- gTTS / ElevenLabs API / Sarvam API (for emotion-based voice synthesis)
- Figma / Blender
- Optional: GPU or high-performance compute system for model training

“This project explores how empathy and emotion can become integral parts of AI-driven learning interfaces—making education more human-centered, responsive, and engaging.”